

$$(i) \quad Ax = b \quad \quad \quad Rx = d$$

$$x = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$$

$$x \rightarrow 3 \times 1$$

$$A \rightarrow n \times 3$$

$$\dim(N(A)) = 2 = n - r$$

$$a) \quad R \rightarrow 3 \times 3$$

$$\dim(N(A)) = 2 = 3 - r$$

$$\boxed{r = 1}$$

$$R \rightarrow \begin{bmatrix} 1 & - & - \\ 0 & - & - \\ 0 & - & - \end{bmatrix}$$

$$d = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}$$

$$R \rightarrow \begin{bmatrix} 1 & -2 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

b)

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ -5 & 0 & 1 \end{bmatrix}$$

$$E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 5 & 0 & 1 \end{bmatrix}$$

$$A = E^{-1} R$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 5 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -2 & -5 \\ 3 & -6 & -15 \\ 5 & -10 & -25 \end{bmatrix}$$

$$b = E^{-1} d \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 5 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 4 \\ 12 \\ 20 \end{bmatrix}$$

② $\rightarrow A^T$ most

~~A~~ =
$$\begin{bmatrix} 0 & 1 & 2 & 2 \\ 0 & 3 & 8 & 7 \\ 0 & 0 & 4 & 2 \end{bmatrix}$$

~~silly mistake~~

g)

$$Ax = 0$$

ref

$$\begin{bmatrix} 0 & 1 & 2 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 4 & 2 \end{bmatrix} \sim$$

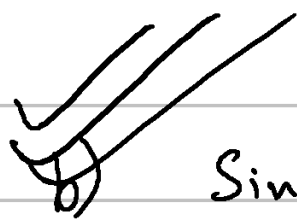
$$\begin{bmatrix} 0 & 1 & 2 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

sp solⁿ $\rightarrow$

$$c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ -1/2 \\ 1 \end{bmatrix}$$

span
of these 2
vectors



Since $\text{rank}(A) = 2$, the column space has 2 vectors in the basis

lets find the 2 independent column vectors in A , 2nd & 3rd for example since they were pivot column

$$C(A) = c_1 \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 8 \\ 4 \end{bmatrix}$$



$$B = \begin{bmatrix} A & A \\ A & A \end{bmatrix}$$



$$\text{ref}(B) = \begin{bmatrix} A & A \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & (1) & 0 & -1 & 0 & 1 & 0 & -1 \\ 0 & 0 & (1) & 1/2 & 0 & 0 & 1 & 1/2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & - & - & - & - & - & - & - \\ 0 & - & - & - & - & - & - & - \end{bmatrix}$$

③

~~a)~~

$$A_{3 \times 5}$$

$$\text{rank}(A) = 3$$

$$Ax = b$$

$$\# \text{ free vars} = 2 \quad \dim N(A) = 2$$

• number of solutions $\rightarrow$ many

• $C(A) \rightarrow$ 3 dimensional $\text{rank} = 3 = \mathbb{R}^3$
 $C(A) \neq \mathbb{R}^5 \quad x \in \mathbb{R}^5$

i) sometimes but not always $\times \times$

ii) has many solutions $\checkmark$

Always because
 $b \in \mathbb{R}^3$ $C(A) = \mathbb{R}^3$

~~4)~~ • The column space of A is $\mathbb{R}^3$

• The nullspace is a 2 dimensional subspace in $\mathbb{R}^5$

④ $A = \begin{bmatrix} 2 & 1 \\ 6 & 5 \\ 2 & 4 \end{bmatrix}$

~~g)~~ $\text{rref}(A) \rightarrow \begin{bmatrix} 2 & 1 \\ 0 & 2 \\ 0 & 3 \end{bmatrix}$

$\begin{bmatrix} 2 & 1 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

only one solution for every b
and vice versa. Not always exists.

Hence if x is a solution to
 $Ax = b$ then b lies in the
column space

$$\text{b)} \quad \begin{bmatrix} 2 & 1 \\ 6 & 5 \\ 2 & 4 \end{bmatrix} x = \begin{bmatrix} 8 \\ 28 \\ 14 \end{bmatrix}$$

$$\rightarrow \left[\begin{array}{cc|c} 2 & 1 & 8 \\ 6 & 5 & 28 \\ 2 & 4 & 14 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cc|c} 2 & 1 & 8 \\ 0 & 2 & 4 \\ 0 & 3 & 6 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 2 & 1 & 8 \\ 0 & 2 & 4 \\ 0 & 0 & 0 \end{array} \right]$$

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix} \rightarrow x_p \text{ is a solution}$$

Yes